

Online Resource 1
Supplementary Information for
Extrinsic Newton Method with Retractions
for Zeros of Vector Fields on Regular Level-Set Submanifolds

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Journal: *Calcolo*

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Contents of Online Resource 1

This online resource complements the main article. It presents an example on the product torus $S^1 \times S^1$ and a nonlinear field on $\text{St}(3, 2)$. It also contains the complete trajectories of residuals and cumulative times for the eight experiments, the moderate-precision timing protocol, the controls for the Riemannian exponential on $SL(2, \mathbb{R})$, and the information required to reproduce the computations.

In all comparisons, the same equation is used to compute the Newton direction, evaluated at the iterate of each variant. The differences arise exclusively from the update by means of the exponential map or the retraction under consideration.

S1 Example on the product torus $S^1 \times S^1$

Consider the product torus

$$\mathcal{M} = S^1 \times S^1 = \{(x, y, z, w) \in \mathbb{R}^4 : x^2 + y^2 = 1, \quad z^2 + w^2 = 1\}. \quad (\text{S1})$$

The two constraints are grouped in the mapping

$$F(x, y, z, w) = \begin{pmatrix} x^2 + y^2 - 1 \\ z^2 + w^2 - 1 \end{pmatrix}. \quad (\text{S2})$$

Its derivative is

$$DF(p) = \begin{pmatrix} 2x & 2y & 0 & 0 \\ 0 & 0 & 2z & 2w \end{pmatrix}. \quad (\text{S3})$$

Since $(x, y) \neq (0, 0)$ and $(z, w) \neq (0, 0)$ on $\mathcal{M}$, the two rows of $DF(p)$ are linearly independent. Therefore, 0 is a regular value and $\mathcal{M}$ is a regular submanifold of dimension two.

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Tangent space, pseudoinverse, and projector

The tangent space is

$$\mathcal{T}_p\mathcal{M} = \left\{ (u_1, u_2, u_3, u_4)^\top : xu_1 + yu_2 = 0, \quad zu_3 + wu_4 = 0 \right\}. \quad (\text{S4})$$

Since

$$\mathbf{D}F(p)\mathbf{D}F(p)^\top = 4I_2,$$

the Moore–Penrose pseudoinverse is

$$\mathbf{D}F(p)^\dagger = \frac{1}{2} \begin{pmatrix} x & 0 \\ y & 0 \\ 0 & z \\ 0 & w \end{pmatrix}. \quad (\text{S5})$$

Therefore, the orthogonal projector onto $\mathcal{T}_p\mathcal{M}$ is

$$\mathbf{P}_p = I_4 - \mathbf{D}F(p)^\dagger \mathbf{D}F(p) = \begin{pmatrix} y^2 & -xy & 0 & 0 \\ -xy & x^2 & 0 & 0 \\ 0 & 0 & w^2 & -zw \\ 0 & 0 & -zw & z^2 \end{pmatrix}. \quad (\text{S6})$$

A natural orthonormal basis of $\mathcal{T}_p\mathcal{M}$ is

$$e_\theta(p) = (-y, x, 0, 0)^\top, \quad e_\varphi(p) = (0, 0, -w, z)^\top. \quad (\text{S7})$$

We write

$$Z_p = \begin{pmatrix} -y & 0 \\ x & 0 \\ 0 & -w \\ 0 & z \end{pmatrix}. \quad (\text{S8})$$

Then

$$Z_p^\top Z_p = I_2, \quad \mathbf{P}_p = Z_p Z_p^\top.$$

Definition of the field

We define

$$f_1(y, w) = y + \frac{1}{2}w + y^2 + \frac{1}{5}yw, \quad (\text{S9})$$

$$f_2(y, w) = -\frac{1}{3}y + w + w^2 + \frac{1}{4}yw. \quad (\text{S10})$$

We consider the ambient field

$$\tilde{X}(x, y, z, w) = f_1(y, w) \begin{pmatrix} -y \\ x \\ 0 \\ 0 \end{pmatrix} + f_2(y, w) \begin{pmatrix} 0 \\ 0 \\ -w \\ z \end{pmatrix}. \quad (\text{S11})$$

In coordinates,

$$\tilde{X}(x, y, z, w) = \begin{pmatrix} -yf_1(y, w) \\ xf_1(y, w) \\ -wf_2(y, w) \\ zf_2(y, w) \end{pmatrix}. \quad (\text{S12})$$

Let us verify tangency. By (S3),

$$\begin{aligned} DF(p)\tilde{X}(p) &= \begin{pmatrix} 2x & 2y & 0 & 0 \\ 0 & 0 & 2z & 2w \end{pmatrix} \begin{pmatrix} -yf_1 \\ xf_1 \\ -wf_2 \\ zf_2 \end{pmatrix} \\ &= \begin{pmatrix} -2xyf_1 + 2xyf_1 \\ -2zwf_2 + 2zwf_2 \end{pmatrix} \\ &= 0. \end{aligned}$$

Therefore,

$$X := \tilde{X}|_{\mathcal{M}}$$

is a tangent vector field on $\mathcal{M}$.

The point

$$p_* = (1, 0, 1, 0)^\top \tag{S13}$$

is a zero, since

$$f_1(0, 0) = f_2(0, 0) = 0.$$

Ambient Jacobian

The derivatives of the coefficients are

$$\begin{aligned} \partial_y f_1 &= 1 + 2y + \frac{1}{5}w, & \partial_w f_1 &= \frac{1}{2} + \frac{1}{5}y, \\ \partial_y f_2 &= -\frac{1}{3} + \frac{1}{4}w, & \partial_w f_2 &= 1 + 2w + \frac{1}{4}y. \end{aligned}$$

For brevity, we write

$$f_{1y} = \partial_y f_1, \quad f_{1w} = \partial_w f_1, \quad f_{2y} = \partial_y f_2, \quad f_{2w} = \partial_w f_2.$$

The ambient Jacobian of (S12) is

$$D\tilde{X}(p) = \begin{pmatrix} 0 & -f_1 - yf_{1y} & 0 & -yf_{1w} \\ f_1 & xf_{1y} & 0 & xf_{1w} \\ 0 & -wf_{2y} & 0 & -f_2 - wf_{2w} \\ 0 & zf_{2y} & f_2 & zf_{2w} \end{pmatrix}. \tag{S14}$$

The extrinsic covariant derivative is

$$\nabla_\eta X(p) = P_p D\tilde{X}(p)(\eta).$$

Reduced Newton system

Let

$$\eta = \alpha e_\theta(p) + \beta e_\varphi(p) = Z_p \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The reduced matrix is

$$H(p) = Z_p^\top P_p D\tilde{X}(p) Z_p.$$

A direct computation gives

$$H(p) = \begin{pmatrix} x \left(1 + 2y + \frac{1}{5}w \right) & z \left(\frac{1}{2} + \frac{1}{5}y \right) \\ x \left(-\frac{1}{3} + \frac{1}{4}w \right) & z \left(1 + 2w + \frac{1}{4}y \right) \end{pmatrix}. \tag{S15}$$

Likewise,

$$Z_p^\top X(p) = \begin{pmatrix} f_1(y, w) \\ f_2(y, w) \end{pmatrix}. \quad (\text{S16})$$

The Newton direction is obtained by solving the reduced system

$$H(p) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = - \begin{pmatrix} f_1(y, w) \\ f_2(y, w) \end{pmatrix}. \quad (\text{S17})$$

If

$$\Delta(p) = \det H(p) \neq 0,$$

then

$$\alpha = \frac{-h_{22}f_1 + h_{12}f_2}{\Delta(p)}, \quad (\text{S18})$$

$$\beta = \frac{h_{21}f_1 - h_{11}f_2}{\Delta(p)}. \quad (\text{S19})$$

At the zero p_* ,

$$H(p_*) = \begin{pmatrix} 1 & \frac{1}{2} \\ -\frac{1}{3} & 1 \end{pmatrix}, \quad (\text{S20})$$

and

$$\det H(p_*) = 1 + \frac{1}{6} = \frac{7}{6} \neq 0. \quad (\text{S21})$$

Therefore, p_* is a regular zero. The inverse function theorem guarantees that it is the unique zero of the field in a sufficiently small neighborhood of p_* .

Angular interpretation

We can parametrize the torus by

$$p(\theta, \varphi) = (\cos \theta, \sin \theta, \cos \varphi, \sin \varphi). \quad (\text{S22})$$

Then

$$e_\theta = \frac{\partial p}{\partial \theta}, \quad e_\varphi = \frac{\partial p}{\partial \varphi}.$$

The field is written as

$$X(p(\theta, \varphi)) = f_1(\sin \theta, \sin \varphi)e_\theta + f_2(\sin \theta, \sin \varphi)e_\varphi.$$

The matrix (S15) is precisely the Jacobian

$$\frac{\partial(f_1, f_2)}{\partial(\theta, \varphi)}.$$

This equality is used in the code only as a verification. The direction is computed with the extrinsic formula

$$Z_p^\top P_p D\tilde{X}(p) Z_p.$$

Exponential map and retraction

Since the torus is the product of two unit circles, the exponential map acts by a rotation in each factor. For

$$\eta = \alpha e_\theta + \beta e_\varphi,$$

we have

$$\begin{aligned} \text{Exp}_p(\eta) = & (x \cos \alpha - y \sin \alpha, y \cos \alpha + x \sin \alpha, \\ & z \cos \beta - w \sin \beta, w \cos \beta + z \sin \beta). \end{aligned} \quad (\text{S23})$$

A closed-form retraction is obtained by normalizing each pair separately:

$$R_p(\eta) = \left(\frac{x - \alpha y}{\sqrt{1 + \alpha^2}}, \frac{y + \alpha x}{\sqrt{1 + \alpha^2}}, \frac{z - \beta w}{\sqrt{1 + \beta^2}}, \frac{w + \beta z}{\sqrt{1 + \beta^2}} \right). \quad (\text{S24})$$

This formula applies the blockwise homogeneous retraction to the two circular constraints. In particular,

$$R_p(0) = p, \quad \text{DR}_p(0) = I_{\text{T}_p \mathcal{M}}.$$

Updated numerical results

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	5.87335×10^{-1}	5.87335×10^{-1}	0.000439	0.000431
1	3.135593×10^{-1}	2.353352×10^{-1}	0.012963	0.005805
2	3.721479×10^{-2}	2.590881×10^{-2}	0.024704	0.010646
3	7.835498×10^{-4}	4.2763×10^{-4}	0.035829	0.015520
4	4.656764×10^{-7}	1.298178×10^{-7}	0.047277	0.021596
5	1.427445×10^{-13}	1.161032×10^{-14}	0.058077	0.026513
6	1.489288×10^{-26}	9.508624×10^{-29}	0.068567	0.031778
7	1.497717×10^{-52}	6.264169×10^{-57}	0.077042	0.037412
8	$1.607123 \times 10^{-104}$	$2.755944 \times 10^{-113}$	0.086658	0.042781
9	1.76902×10^{-208}	$5.279411 \times 10^{-226}$	0.095481	0.048186
10	$2.217477 \times 10^{-416}$	$1.952676 \times 10^{-451}$	0.102580	0.053230
11	$3.395322 \times 10^{-832}$	$2.655362 \times 10^{-902}$	0.110673	0.058621

Table S1: Product torus: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

S2 The Stiefel manifold: costly exponential map and simple retraction

Consider the Stiefel manifold

$$\text{St}(n, r) = \left\{ Y \in \mathbb{R}^{n \times r} : Y^\top Y = I_r \right\}.$$

This manifold is the regular level set of the mapping

$$F(Y) = Y^\top Y - I_r,$$

understood as taking values in the space $\text{Sym}(r)$ of symmetric matrices. Its derivative is

$$\text{DF}(Y)(H) = Y^\top H + H^\top Y. \quad (\text{S25})$$

This mapping is surjective. Indeed, if $S \in \text{Sym}(r)$, by taking

$$H = \frac{1}{2} Y S$$

we obtain

$$DF(Y)(H) = \frac{1}{2}S + \frac{1}{2}S = S.$$

Therefore, 0 is a regular value of F .

The tangent space is

$$T_Y \text{St}(n, r) = \left\{ \eta \in \mathbb{R}^{n \times r} : Y^\top \eta + \eta^\top Y = 0 \right\}. \quad (\text{S26})$$

In this subsection, we use the Euclidean metric induced by the Frobenius inner product,

$$\langle U, V \rangle_F = \text{tr}(U^\top V).$$

The normal space is

$$N_Y \text{St}(n, r) = \{YS : S \in \text{Sym}(r)\},$$

and the orthogonal projector onto the tangent space is

$$P_Y(Z) = Z - Y \text{sym}(Y^\top Z), \quad (\text{S27})$$

where

$$\text{sym}(B) = \frac{1}{2}(B + B^\top).$$

Riemannian exponential for the Euclidean metric

The Riemannian exponential of the induced Euclidean metric can be computed by means of a block formula. Let

$$\eta \in T_Y \text{St}(n, r),$$

and define

$$A = Y^\top \eta, \quad S = \eta^\top \eta.$$

Since η is tangent,

$$A^\top = -A.$$

The Euclidean geodesic with initial conditions

$$\gamma(0) = Y, \quad \dot{\gamma}(0) = \eta$$

satisfies

$$\text{Exp}_Y^E(\eta) = \begin{bmatrix} Y & \eta \end{bmatrix} \exp \begin{pmatrix} A & -S \\ I_r & A \end{pmatrix} \begin{pmatrix} \exp(-A) \\ 0 \end{pmatrix}. \quad (\text{S28})$$

This expression corresponds to the embedded, or Euclidean, metric and requires, in general, a matrix exponential of size $2r \times 2r$, in addition to the exponential of the skew-symmetric matrix $A \in \mathbb{R}^{r \times r}$.

In the case $\text{St}(3, 2)$, formula (S28) requires computing one matrix exponential of size 4×4 and another of size 2×2 .

Polar retraction

A much simpler alternative is the polar retraction

$$R_Y^{\text{pol}}(\eta) = (Y + \eta)(I_r + \eta^\top \eta)^{-1/2}. \quad (\text{S29})$$

Indeed, using the tangency condition,

$$\begin{aligned} (Y + \eta)^\top (Y + \eta) &= Y^\top Y + Y^\top \eta + \eta^\top Y + \eta^\top \eta \\ &= I_r + \eta^\top \eta. \end{aligned}$$

Therefore,

$$\begin{aligned} (R_Y^{\text{pol}}(\eta))^\top R_Y^{\text{pol}}(\eta) &= (I_r + \eta^\top \eta)^{-1/2} (I_r + \eta^\top \eta) (I_r + \eta^\top \eta)^{-1/2} \\ &= I_r. \end{aligned}$$

Thus,

$$R_Y^{\text{pol}}(\eta) \in \text{St}(n, r).$$

Moreover,

$$R_Y^{\text{pol}}(0) = Y, \quad DR_Y^{\text{pol}}(0)(\eta) = \eta.$$

Therefore, it is a retraction and, in particular, it satisfies the first-order assumptions used in the convergence theorems.

Its geometric cost is concentrated in the inverse square root of an $r \times r$ matrix. For $\text{St}(3, 2)$, only the inverse square root of a symmetric positive-definite matrix of size 2×2 is required.

A nonlinear and coupled field on $\text{St}(3, 2)$

From now on, consider

$$Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ y_{31} & y_{32} \end{pmatrix} \in \text{St}(3, 2).$$

We first define three linear forms:

$$u_1(Y) = y_{11} + \frac{2}{5}y_{22} - \frac{3}{10}y_{31} + \frac{1}{5}y_{12}, \quad (\text{S30})$$

$$u_2(Y) = y_{21} - \frac{1}{5}y_{12} + \frac{1}{2}y_{32} + \frac{1}{10}y_{11}, \quad (\text{S31})$$

$$u_3(Y) = y_{31} + \frac{3}{10}y_{22} - \frac{2}{5}y_{21} + \frac{1}{5}y_{32}. \quad (\text{S32})$$

From them, we construct the nonlinear functions

$$g_1(Y) = u_1 + u_2^2 - 1.6119, \quad (\text{S33})$$

$$g_2(Y) = u_2 + u_1 u_3 - 1.2468, \quad (\text{S34})$$

$$g_3(Y) = u_3 + u_1^2 + \frac{1}{2}u_2 u_3 - 1.6423. \quad (\text{S35})$$

These expressions depend on the six entries of Y and are coupled. The decimal constants are fixed as part of the definition of the problem; they are not adjusted after running the method. Their choice produces a nontrivial zero, away from the coordinate frames, and a regular reduced matrix.

For

$$v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix},$$

denote by

$$\widehat{v} = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}$$

the associated skew-symmetric matrix. We define

$$X(Y) = \widehat{g(Y)} Y, \quad g(Y) = \begin{pmatrix} g_1(Y) \\ g_2(Y) \\ g_3(Y) \end{pmatrix}. \quad (\text{S36})$$

The field is tangent. Indeed,

$$\begin{aligned} Y^\top X(Y) + X(Y)^\top Y &= Y^\top \widehat{g(Y)} Y + Y^\top \widehat{g(Y)}^\top Y \\ &= Y^\top \left(\widehat{g(Y)} - \widehat{g(Y)}^\top \right) Y \\ &= 0. \end{aligned}$$

Moreover,

$$X(Y) = 0 \iff g(Y) = 0.$$

Indeed, if $\widehat{g(Y)} \neq 0$, its null space in $\mathbb{R}^3$ has dimension one and cannot contain the two linearly independent columns of $Y \in \text{St}(3, 2)$.

Ambient derivative and covariant derivative

Let

$$H = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \\ h_{31} & h_{32} \end{pmatrix}.$$

The derivatives of the linear forms are

$$\begin{aligned} Du_1(Y)(H) &= h_{11} + \frac{2}{5}h_{22} - \frac{3}{10}h_{31} + \frac{1}{5}h_{12}, \\ Du_2(Y)(H) &= h_{21} - \frac{1}{5}h_{12} + \frac{1}{2}h_{32} + \frac{1}{10}h_{11}, \\ Du_3(Y)(H) &= h_{31} + \frac{3}{10}h_{22} - \frac{2}{5}h_{21} + \frac{1}{5}h_{32}. \end{aligned}$$

If we write

$$Du(Y)(H) = \begin{pmatrix} \delta u_1 \\ \delta u_2 \\ \delta u_3 \end{pmatrix},$$

then

$$Dg(Y)(H) = \begin{pmatrix} 1 & 2u_2 & 0 \\ u_3 & 1 & u_1 \\ 2u_1 & \frac{1}{2}u_3 & 1 + \frac{1}{2}u_2 \end{pmatrix} Du(Y)(H). \quad (\text{S37})$$

By the product rule,

$$D\tilde{X}(Y)(H) = D\widehat{g(Y)}(H) Y + \widehat{g(Y)} H. \quad (\text{S38})$$

The covariant derivative for the induced Euclidean metric is

$$\nabla_H X(Y) = P_Y \left(D\widehat{g(Y)}(H) Y + \widehat{g(Y)} H \right). \quad (\text{S39})$$

Reduced 3×3 system

Consider the matrices

$$\begin{aligned} K_1 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, & K_2 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

We define

$$B_i(Y) = K_i Y, \quad i = 1, 2, 3. \quad (\text{S40})$$

Since $\text{St}(3, 2)$ has dimension three, these vectors form a global basis of $T_Y \text{St}(3, 2)$.

We write

$$\eta = \xi_1 B_1(Y) + \xi_2 B_2(Y) + \xi_3 B_3(Y).$$

The Newton equation reduces to the system

$$M(Y) \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = -b(Y), \quad (\text{S41})$$

where

$$M_{ij}(Y) = \langle B_i(Y), \nabla_{B_j(Y)} X(Y) \rangle_F,$$

and

$$b_i(Y) = \langle B_i(Y), X(Y) \rangle_F.$$

Once (S41) has been solved,

$$\eta = \sum_{j=1}^3 \xi_j B_j(Y).$$

Updated numerical results

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	9.758255×10^{-1}	9.758255×10^{-1}	0.001169	0.001196
1	8.201328×10^{-1}	6.478493×10^{-1}	0.088819	0.024914
2	1.077263×10^{-1}	1.037519×10^{-1}	0.165888	0.047424
3	2.286715×10^{-2}	1.882023×10^{-2}	0.233804	0.072950
4	1.184656×10^{-3}	7.663722×10^{-4}	0.297837	0.094344
5	3.656274×10^{-6}	1.572548×10^{-6}	0.358700	0.115600
6	3.899457×10^{-11}	7.277254×10^{-12}	0.407913	0.137056
7	4.471085×10^{-21}	1.555223×10^{-22}	0.449970	0.158501
8	5.845408×10^{-41}	7.069257×10^{-44}	0.486054	0.180165
9	9.986523×10^{-81}	1.460687×10^{-86}	0.520960	0.207514
10	$2.915642 \times 10^{-160}$	$6.237787 \times 10^{-172}$	0.556403	0.229489
11	$2.485343 \times 10^{-319}$	1.13757×10^{-342}	0.592360	0.250823
12	$1.805863 \times 10^{-637}$	$3.783271 \times 10^{-684}$	0.626350	0.271646

Table S2: Nonlinear Stiefel problem: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

S3 Complete multiprecision trajectories

The multiprecision protocol uses 1200 decimal digits, tolerance 10^{-500} , and three alternating repetitions of each variant. In each row, the residual and cumulative time belong to the same protocol. The times are the cumulative medians by iteration. If one variant terminates earlier, no subsequent iterations are invented; for this reason, dashes appear for the retraction in the example on S^3 after its one-step termination.

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	7.874643×10^{-1}	7.874643×10^{-1}	0.000268	0.000256
1	1.434361×10^{-1}	1.436173×10^{-1}	0.006942	0.003726
2	1.267638×10^{-2}	1.267914×10^{-2}	0.013534	0.006721
3	1.524631×10^{-4}	1.524634×10^{-4}	0.019511	0.009878
4	2.323057×10^{-8}	2.323057×10^{-8}	0.025137	0.013068
5	5.396595×10^{-16}	5.396595×10^{-16}	0.030297	0.016241
6	2.912324×10^{-31}	2.912324×10^{-31}	0.034642	0.019385
7	8.481631×10^{-62}	8.481631×10^{-62}	0.038993	0.022638
8	$7.193807 \times 10^{-123}$	$7.193807 \times 10^{-123}$	0.043338	0.025694
9	$5.175086 \times 10^{-245}$	$5.175086 \times 10^{-245}$	0.047615	0.028676
10	$2.678152 \times 10^{-489}$	$2.678152 \times 10^{-489}$	0.051941	0.031653
11	$7.172496 \times 10^{-978}$	$7.172496 \times 10^{-978}$	0.057418	0.034571

Table S3: Cylinder: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	5.0×10^{-1}	5.0×10^{-1}	0.001096	0.000922
1	5.372561×10^{-2}	$3.363705 \times 10^{-1202}$	0.016217	0.014422
2	5.182658×10^{-5}	—	0.029973	—
3	4.640197×10^{-14}	—	0.043548	—
4	3.330336×10^{-41}	—	0.057234	—
5	1.23124×10^{-122}	—	0.071222	—
6	$6.221675 \times 10^{-367}$	—	0.084759	—
7	$8.027875 \times 10^{-1100}$	—	0.098664	—

Table S4: S^3 : complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	5.484387×10^{-1}	5.484387×10^{-1}	0.000672	0.000626
1	3.607681×10^{-1}	3.256465×10^{-1}	2.021257	0.016090
2	2.326791×10^{-2}	3.606411×10^{-2}	2.977476	0.029715
3	4.350643×10^{-4}	3.564352×10^{-4}	3.207198	0.042812
4	3.951101×10^{-8}	2.632585×10^{-8}	3.399987	0.056178
5	4.043158×10^{-16}	1.476852×10^{-16}	3.598228	0.069326
6	3.034939×10^{-32}	4.713945×10^{-33}	3.790723	0.082565
7	1.551148×10^{-64}	4.778443×10^{-66}	4.006702	0.096186
8	$4.170436 \times 10^{-129}$	$4.861653 \times 10^{-132}$	4.198194	0.110084
9	$3.053869 \times 10^{-258}$	5.01734×10^{-264}	4.394884	0.124258
10	$1.636597 \times 10^{-516}$	$5.364794 \times 10^{-528}$	4.586303	0.137583

Table S5: $SL(2, \mathbb{R})$: Riemannian exponential and homogeneous retraction. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	2.606035×10^{-1}	2.606035×10^{-1}	0.001357	0.001476
1	1.762323×10^{-2}	1.28056×10^{-2}	0.016806	0.013220
2	5.149984×10^{-6}	1.537462×10^{-6}	0.031659	0.025334
3	1.419023×10^{-16}	3.006613×10^{-18}	0.045072	0.037162
4	3.218826×10^{-48}	2.446631×10^{-53}	0.058293	0.049224
5	$3.986347 \times 10^{-143}$	1.38595×10^{-158}	0.071708	0.061453
6	$7.883916 \times 10^{-428}$	$2.588342 \times 10^{-474}$	0.084551	0.073689
7	$4.119681 \times 10^{-1201}$	$2.464168 \times 10^{-1201}$	0.098081	0.085713

Table S6: Wahba problem: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	2.339125×10^0	2.339125×10^0	0.004200	0.003970
1	3.810634×10^{-1}	1.492113×10^{-1}	0.036781	0.030213
2	1.49713×10^{-3}	2.94527×10^{-5}	0.068347	0.060097
3	1.441875×10^{-10}	2.597738×10^{-16}	0.099895	0.085820
4	1.307331×10^{-31}	1.550253×10^{-48}	0.128240	0.110914
5	9.642509×10^{-95}	5.45052×10^{-145}	0.156681	0.136953
6	$3.898301 \times 10^{-284}$	2.61254×10^{-434}	0.186891	0.163861
7	2.56216×10^{-852}	$1.254982 \times 10^{-1201}$	0.215074	0.188965

Table S7: Toroidal von Mises model: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

k	$\ X(p_k)\ _{\text{Exp}}$	$\ X(p_k)\ _R$	$t_{\text{Exp},k}$ (s)	$t_{R,k}$ (s)
0	4.719654×10^{-1}	4.719654×10^{-1}	0.002236	0.002561
1	5.353999×10^{-2}	4.625133×10^{-2}	0.086563	0.032679
2	5.624617×10^{-4}	4.930562×10^{-4}	0.160845	0.063338
3	4.173036×10^{-8}	2.588569×10^{-8}	0.222026	0.088974
4	3.18385×10^{-16}	1.457545×10^{-16}	0.269569	0.115542
5	1.33614×10^{-32}	2.260606×10^{-33}	0.317738	0.144748
6	3.264013×10^{-65}	1.11161×10^{-66}	0.357374	0.173028
7	$1.404269 \times 10^{-130}$	$1.314879 \times 10^{-133}$	0.398799	0.200872
8	$3.605363 \times 10^{-261}$	$3.760746 \times 10^{-267}$	0.439157	0.230718
9	$1.713344 \times 10^{-522}$	$1.504975 \times 10^{-534}$	0.479302	0.256425

Table S8: Brockett eigenproblem: complete trajectory of residuals and cumulative times. The times are cumulative medians of three alternating runs with 1200 decimal digits and tolerance 10^{-500} .

S4 Timing summary and moderate-precision protocol

Problem	k_{Exp}	k_R	t_{Exp}	MAD_{Exp}	t_R	MAD_R	Reduction
Cylinder	11	11	0.057418	0.000068	0.034571	0.000360	39.8%
Sphere S^3	7	1	0.098664	0.000530	0.014422	0.000166	85.4%
Product torus	11	11	0.110673	0.003233	0.058621	0.001686	47.0%
Nonlinear $SL(2, \mathbb{R})$	10	10	4.586303	0.093026	0.137583	0.003157	97.0%
Nonlinear Stiefel problem	12	12	0.626350	0.003254	0.271646	0.006032	56.6%
Wahba on S^3	7	7	0.098081	0.003024	0.085713	0.001819	12.6%
Toroidal von Mises	7	7	0.215074	0.003186	0.188965	0.011475	12.1%
Brockett on $\text{St}(3, 2)$	9	9	0.479302	0.013373	0.256425	0.010924	46.5%

Table S9: Summary of the multiprecision protocol. Times are expressed in seconds.

Table S9 answers the question of how much it costs to reach the extreme tolerance used to study the asymptotic regime. It should not be interpreted as a universal benchmark: the cost of transcendental functions, matrix exponentials, and geodesic integration increases with the requested precision.

As an additional control, a second protocol was run with 50 digits and tolerance 10^{-14} . Each method was warmed up twice and measured over eleven repetitions, alternating the order of execution. The median and the median absolute deviation (MAD) of the total time are reported.

Problem	k_{Exp}	k_R	t_{Exp}	MAD_{Exp}	t_R	MAD_R	Reduction
Cylinder	5	5	0.004506	0.000485	0.003922	0.000095	13.0%
Sphere S^3	4	1	0.010709	0.000462	0.002558	0.000058	76.1%
Product torus	6	6	0.009887	0.000843	0.008421	0.000281	14.8%
Nonlinear $SL(2, \mathbb{R})$	5	5	0.172084	0.005870	0.015709	0.000640	90.9%
Nonlinear Stiefel problem	7	7	0.057641	0.000564	0.033332	0.000638	42.2%
Wahba on S^3	3	3	0.008871	0.000848	0.008168	0.000637	7.9%
Toroidal von Mises	4	3	0.005742	0.000076	0.004185	0.000081	27.1%
Brockett on $\text{St}(3, 2)$	4	4	0.042311	0.001870	0.027628	0.001075	34.7%

Table S10: Total times at moderate precision: 50 digits, tolerance 10^{-14} , two warm-up runs, and eleven alternating repetitions.

The clearest reductions in both protocols appear for $SL(2, \mathbb{R})$ and in the Stiefel problems. In the low-cost experiments, the differences should be interpreted together with the MAD and not as a universal superiority.

S5 Controls for the Riemannian exponential on $SL(2, \mathbb{R})$

The Riemannian exponential is evaluated by integrating the geodesic equation of the induced Frobenius metric. The implementation uses a recurrent Taylor series and adapts the tolerance to the residual of the Newton step. For each geodesic, the truncation order, an a posteriori estimate based on the recent ratios of the coefficients, and controls of feasibility, tangency, the geodesic equation, and energy conservation are recorded.

Step	Residual before the step	Requested tolerance	Series order	Tail estimate	Recent ratio
0	5.4844×10^{-1}	1.0×10^{-80}	109	9.007×10^{-81}	1.96×10^{-1}
1	3.6077×10^{-1}	1.0×10^{-80}	76	3.078×10^{-81}	9.795×10^{-2}
2	2.3268×10^{-2}	1.0×10^{-80}	36	6.739×10^{-81}	6.723×10^{-3}
3	4.3506×10^{-4}	1.0×10^{-80}	33	2.855×10^{-136}	9.931×10^{-5}
4	3.9511×10^{-8}	1.0×10^{-80}	33	2.528×10^{-271}	7.773×10^{-9}
5	4.0432×10^{-16}	1.0×10^{-80}	33	1.492×10^{-536}	7.196×10^{-17}
6	3.0349×10^{-32}	2.795×10^{-95}	33	7.279×10^{-1067}	5.971×10^{-33}
7	1.5511×10^{-64}	3.732×10^{-192}	33	1.081×10^{-2131}	3.324×10^{-65}
8	4.1704×10^{-129}	7.253×10^{-386}	33	9.368×10^{-4263}	8.618×10^{-130}
9	3.0539×10^{-258}	1.0×10^{-560}	33	2.075×10^{-8524}	6.373×10^{-259}

Table S11: Data for the a posteriori truncation criterion for the geodesics of $SL(2, \mathbb{R})$. The tail estimate is not presented as a rigorous certification of the entire future series.

Step	Raw det defect	Stabilization	Tangency defect	Geodesic residual	Energy defect
0	5.979×10^{-82}	4.997×10^{-82}	6.568×10^{-80}	1.687×10^{-77}	6.34×10^{-80}
1	2.843×10^{-82}	2.354×10^{-82}	2.188×10^{-80}	1.54×10^{-78}	2.773×10^{-81}
2	5.1×10^{-83}	4.217×10^{-83}	1.887×10^{-81}	5.889×10^{-80}	5.844×10^{-84}
3	4.275×10^{-141}	3.535×10^{-141}	1.454×10^{-139}	2.686×10^{-137}	1.624×10^{-142}
4	2.85×10^{-279}	2.357×10^{-279}	9.691×10^{-278}	2.155×10^{-276}	2.088×10^{-286}
5	1.663×10^{-552}	1.375×10^{-552}	5.655×10^{-551}	1.194×10^{-549}	1.187×10^{-567}
6	6.523×10^{-1099}	5.393×10^{-1099}	2.218×10^{-1097}	4.927×10^{-1096}	6.718×10^{-1130}
7	1.554×10^{-1201}	0	8.852×10^{-1266}	1.096×10^{-1330}	0
8	7.768×10^{-1202}	1.79×10^{-1201}	1.681×10^{-1330}	4.526×10^{-1460}	8.084×10^{-1460}
9	1.554×10^{-1201}	0	8.084×10^{-1460}	4.835×10^{-1718}	8.412×10^{-1718}

Table S12: Geometric and dynamical controls for the integration of the Riemannian exponential on $SL(2, \mathbb{R})$.

The final normalization of the determinant is used only to correct the rounding error of the integration. Its norm is recorded in [Table S12](#). The code does not use the group exponential.

S6 Implementation and reproducibility

The repository contains eight standalone experiment programs and the global result generator `src/generate_article_tables.py`. The generator accepts the option `--output-root` and creates, under the selected destination, the directories `multiprecision`, `moderate_precision`, and `diagnostics`. It also writes the file `execution_environment.json`.

The reference execution archived in the repository was carried out with:

- Python: 3.13.5;
- mpmath: 1.3.0;
- platform: Linux-4.4.0-x86_64-with-glibc2.41;
- timer: `time.perf_counter`;
- UTC date: 2026-07-16T00:18:43.319220+00:00.

From the repository root, all numerical results are regenerated by

```
python src/generate_article_tables.py --output-root results_local
```

This command writes a local regeneration to `results_local` and leaves the committed reference data in `results` unchanged. The exact numerical dependency is pinned in `requirements.txt`. Detailed execution instructions are provided in `docs/REPRODUCIBILITY.md`, and the two numerical protocols are documented in `docs/NUMERICAL_PROTOCOLS.md`. The committed reference files are stored in `results/multiprecision`, `results/moderate_precision`, and `results/diagnostics`; the run-time metadata are stored in `results/execution_environment.json`. Repository integrity can be checked with `scripts/verify_repository.py`, while the SHA-256 manifest is available at `checksums/SHA256SUMS.txt`.

The programs verify, depending on the problem, tangency of the direction, the residual of the reduced system, satisfaction of the constraints, coincidence of the final roots, and regularity of the zero. No number in the manuscript or Online Resource 1 is adjusted manually.